University of Calgary
Faculty of Science Midterm
PHYSICS 365/369

November 7, 2023, 12:30 PM- 1:45 PM Time: 75 minutes

This is a closed-book exam containing a total of 16 equally weighted questions. Please answer all questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the multiple-choice exam questions and the formula sheet (last page). THIS BOOKLET WILL **NOT** BE HANDED IN.

2. A BUBBLE SHEET used to answer multiple-choice questions. **IT WILL BE HANDED IN**.

IMPORTANT: START by entering your ID NUMBER, NAME, and COURSE ID on the bubble sheet.

3. A CALCULATOR. Any scientific or graphing calculator (including a programmable one) may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet. If you are using a PEN, make sure it is BLACK or BLUE in order to ensure the scanner reads your answers properly. PENCILS can also be used, but make sure you press firmly when answering so the scanner reads your answer properly.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

6. Students remaining after **70 minutes** must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

1. These two waves travel along the same string:

$$y_1(x,t) = \left(\sqrt{2}\ \text{mm}\right)\sin(2\pi x - 200\pi t)$$

$$y_2(x,t) = (4\ \text{mm})\sin\left(2\pi x - 200\pi t + \frac{3\pi}{4}\ \text{rad}\right)$$

What is the amplitude of the resultant wave?

A) 3.16 mm
B) 9.0 mm
C) 3.0 mm
D) 10 mm

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2. The figure 1 is a composite of three snapshots, each of a wave traveling along a particular string. The phases for the waves are given by (a) 2x − 4t, (b) 4x − 8t, and (c) 8x − 16t. Which phase corresponds to which wave in the figure?

A) (a) to 1; (b) to 2; (c) to 3
B) (a) to 2; (b) to 3; (c) to 1
C) (a) to 3; (b) to 2; (c) to 1
D) (a) to 3; (b) to 1; (c) to 2

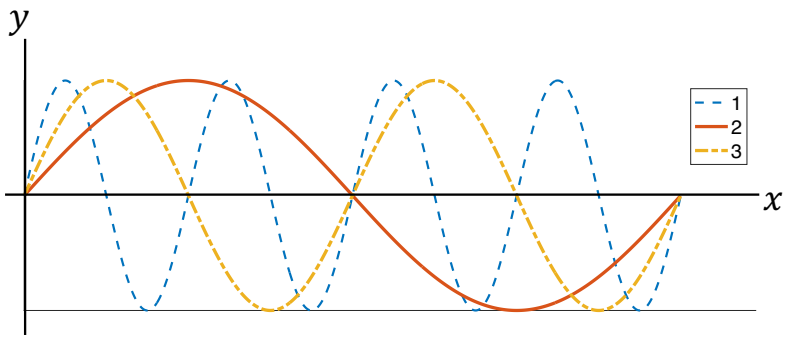
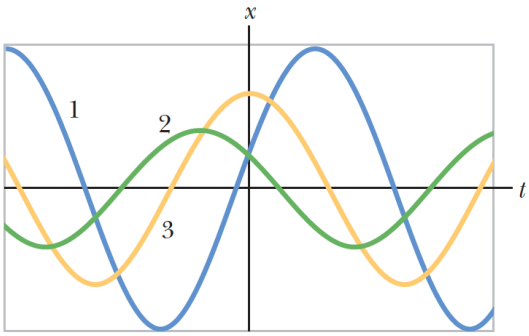


Figure 1

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3. Figure 2 displays the x(t) curves for three experiments involving a specific spring-mass system oscillating in simple harmonic motion (SHM). Rank the curves according to the system's angular frequency and choose the correct answer.

A) $\omega_1 = \omega_2 = \omega_3$
B) $\omega_1 = \omega_2 < \omega_3$
C) $\omega_1 = \omega_2 > \omega_3$
D) $\omega_1 > \omega_3 > \omega_2$

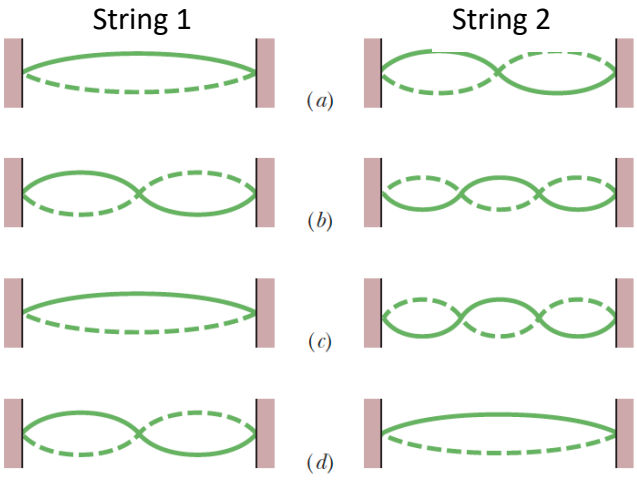


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Figure 2

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4. Strings 1 and 2 have identical lengths and linear densities, but string 2 is under greater tension than string 1. Figure 3 shows four situation, (a) through (d), in which standing wave patterns exist on the two strings. In which situation is there the possibility that strings 1 and 2 are oscillating at the same resonant frequency?

A) (a)
B) (b)
C) (c)
D) (d)



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Figure 3

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5. Two waves are traveling along a string in figure 4. The graph shows the position of the waves at time $t = 0.0$ s. One wave with a maximum amplitude of 0.5 cm is traveling toward the right at 0.5 cm/s. The second wave with a maximum amplitude of 2.0 cm is traveling toward the left at 2.0 cm/s. At what elapsed time will the two waves completely overlap and what will the maximum amplitude be at that time?

A) 2.0 s, 1.5 cm
B) 1.6 s, 2.5 cm
C) 1.0 s, 1.5 cm
D) 1.0 s, 2.5 cm

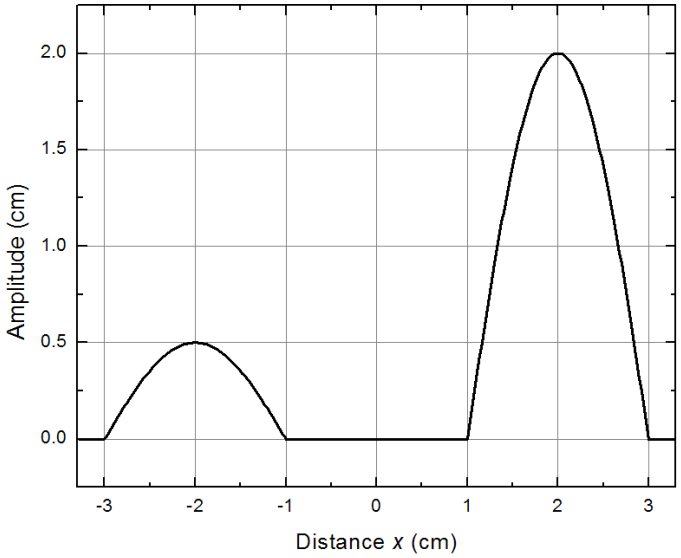


Figure 4

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6. A tsunami is a fast-moving ocean wave train that is produced during an earthquake. Consider such a wave initiated at center of the earthquake off the western coast of South America that reaches the Hawaiian Islands within 15 hours. Which one of the following statements concerning the tsunami is correct?

A) The tsunami carried water from the earthquake center to Hawaii, but it did not carry energy to Hawaii from South America.
B) The tsunami carried energy and water from the earthquake center to Hawaii.
C) The tsunami carried energy from the earthquake center to Hawaii, but it did not carry water to Hawaii from South America.
D) The tsunami did not carry energy or water from the earthquake center to Hawaii.

7. A child is swinging back and forth with a constant period and amplitude (figure 5). Somewhere in front of the child, a stationary horn is emitting a constant tone of frequency f_s. Four points are labeled in the drawing to indicate positions along the arc as the child swings. At which position(s) will the child hear the lowest frequency for the sound from the whistle?

A) at **2** when moving toward **1**
B) at **2** when moving toward **3**
C) at **3** when moving toward **2**
D) at **3** when moving toward **4**

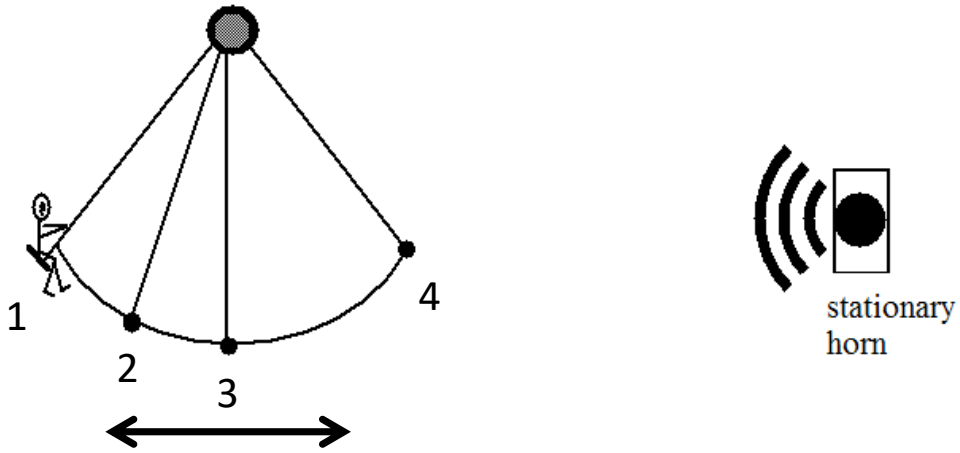


Figure 5

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8. The acceleration $a(t)$ of a particle undergoing SHM is graphed in the Figure 6. The amplitude of the oscillation is $+x_m$. At point 5, particle is at:

A) $+x_m$
B) $-x_m$
C) Between $-x_m$ and 0
D) Between 0 and $+x_m$

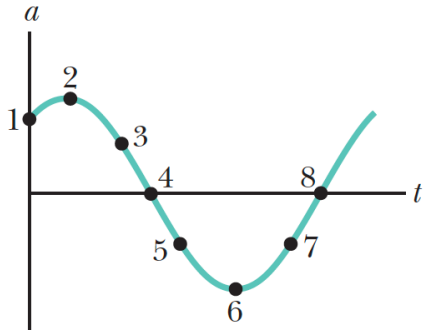


Figure 6

9. Figure 7 shows three pipes, of lengths L and $2L$, in which the first harmonic has been set up. Rank them according to the frequency of the first harmonic, greatest first.

A) 1, 2, 3
B) 1, 3, 2
C) 2, 1, 3
D) 3, 2, 1

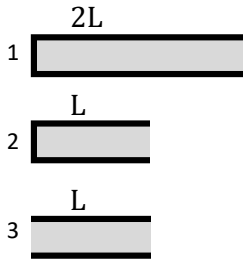


Figure 7

Questions 11, 12 and 13 are related to the following situation:

A sinusoidal wave moving along a string is shown twice in figure 8, as crest A travels in the positive direction of an x axis by distance $d = 6.0$ cm in 4.0 ms. The tick marks along the axis are separated by 10 cm; height $H = 6.00$ mm.

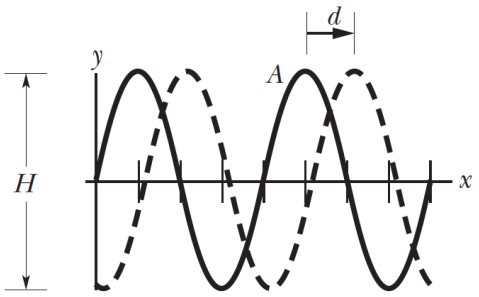


Figure 8

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10. What is the amplitude of the wave?

A) 0.006 m
B) 0.060 m
C) 0.003 m
D) 0.006 m

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11. What is the speed of the wave?

A) 15 m/s
B) 1.5 m/s
C) 0.15 m/s
D) 150 m/s

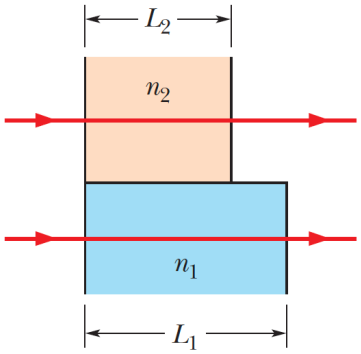
12. What is the angular wave number k?

A) 16 rad/m
B) 1.6 rad/m
C) 0.16 rad/m
D) 160 rad/m

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13. Two waves of light in air, of wavelength λ = 800.0 nm, are initially in phase. They then both travel through a layer of plastic as shown in Figure 9, with L_1 = 4.00 μm, L_2 = 3.50 μm, n_1 = 1.40, and n_2 = 1.60. What multiple of λ gives their phase difference after they both have emerged from the layers?

A) 0.833
B) 0.625
C) 0.714
D) 0.572



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Figure 9

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14. The distance between the first and sixth minima of a single- slit diffraction pattern is 0.35 mm with the screen 40 cm away from the slit, when light of wavelength 550 nm is used. What is the slit width?

A) 2.50 mm
B) 3.14 mm
C) 1.25 mm
D) 4.75 mm

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15. In Figure 10, two radiofrequency point sources S_1 and S_2, separated by distance d = 4.0 m, are radiating in phase with λ = 0.50 m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

A) 16
B) 32
C) 8
D) 36

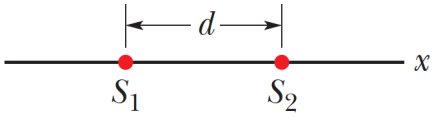


Figure 10

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16. In the Figure 11, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3 (the light does not reflect inside material 2) and r_4 (the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.
(The visible wavelengths range from 0.4μm to 0.7μm)

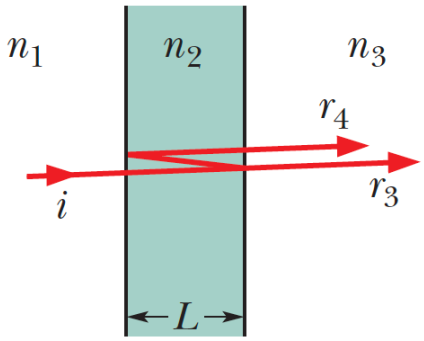


Figure 11

What is the wavelength in the visible range for the values in table below?

n_1	n_2	n_3	Type	L	λ
1.32	1.75	1.39	Min	325	

A) 455 nm
B) 423 nm
C) 478 nm
D) 497 nm

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17. A string having linear mass density μ_1 is joined to a second, lighter string with mass density μ_2 and placed under tension τ. A sinusoidal wave carrying power P is launched into string 1, resulting in a wave in string 2 carrying power $\frac{9}{25}P$. What is μ_2/μ_1?

A) $\frac{3}{5}$
B) $\frac{5}{3}$
C) 8
D) $\frac{1}{8}$

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18. If you were listening to an intense conversation at 80 dB, and you move to a location where the sound intensity decreases by a factor of 100, what would be the new sound intensity level?

A) 70 dB
B) 60 dB
C) 50 dB
D) 40 dB

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19. A tube, which remains open at both ends, sustains a standing wave with harmonics occurring at 200 Hz and 300 Hz, and no harmonics in between. What is the fundamental frequency of this tube?

A) 250 Hz
B) 200 Hz
C) 100 Hz
D) 50 Hz

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20. The figure 12 below shows the intensity of 500 nm light projected through two narrow slits onto a distant screen. The labels on the *x* axis indicate the fringe order of the 2-slit pattern. If the two slits are 2.0 mm apart, what is the width of each slit?

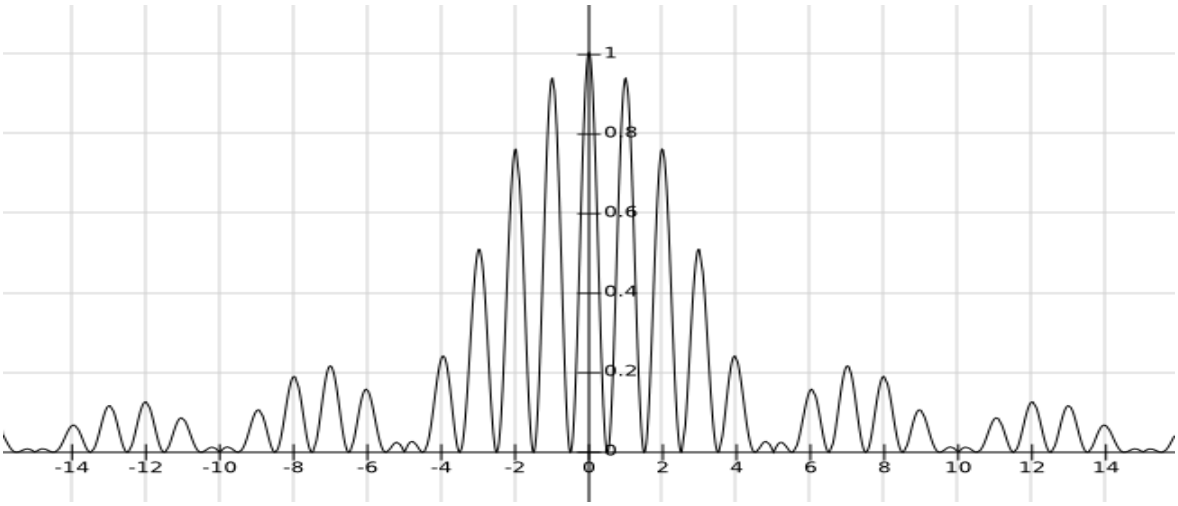


Figure 12

A) 1.0 mm
B) 2.0 mm
C) 2.5 mm
D) 0.4 mm

PHYS 365/369 Formula Sheet

OSCILLATIONS			
$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{L}}$	$T = 2\pi\sqrt{\frac{l}{g}}$
WAVES			
$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \sin(kx \pm \omega t + \phi)$		
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_{vacuuum}}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v\omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v\omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$R^2 + \frac{Z_2}{Z_1}T^2 = 1$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$f = \frac{v}{\lambda} = nf_1$	$\frac{\partial^2 y}{\partial^2 t} = v^2\frac{\partial^2 y}{\partial x^2}$	$\Delta p = -B\frac{\partial s}{\partial x}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$

$x(t) = x_m \cos(\omega t + \Phi)$		$s = s_m \cos(kx - \omega t)$
$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\Phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\Phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$
$\Phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2,\ \ldots.$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5,\ \ldots.$
Pipe open at both ends $f = \frac{v}{\lambda} = \frac{nv}{2L},\ n = 1,\ 2,\ 3, \ldots$		Pipe closed at one end and open at the other $f = \frac{v}{\lambda} = \frac{nv}{4L},\ n = 1,\ 3,\ 5, \ldots$

$d\ \sin(\theta) = m\lambda$ for m= 0, 1, 2, … (Maxima- bright fringes)		
$d\ \sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for m= 0, 1, 2, … (minima—dark fringes),		
$I = 4I_0 \cos^2\left(\frac{1}{2}\Phi\right)$	$\Phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$I = I_0 \cos^2(\theta)$

Reflection	Reflection phase shift
Off lower index	0
Off higher index	0.5 wavelength

$a\sin(\theta) = m\lambda$ for m= 1, 2, 3, …. (minima)

CONSTANTS

$c = 3.00 \times 10^8 \frac{m}{s}$	$R = 8.314 \frac{J}{k.mol}$	$M_{air} = 28.8\frac{g}{mol}$	$g = 9.81 \frac{m}{s}$
$v_{sound\ at\ 20℃ = 343 \frac{m}{s}}$	$T_K = T_{℃} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	$R = r^2$
$I_0 = 1 \times 10^{-12} \frac{W}{m^2}$	$1\ atm = 1.01 \times 10^5 Pa$	$\gamma_{diaatomic} = \frac{7}{5}$	$T = \frac{Z_2}{Z_1}t^2$

MATHEMATICAL FORMULAE

$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$	$\sin(\alpha) \pm \sin(\beta) = 2\sin\frac{(\alpha \pm \beta)}{2}\cos\frac{(\alpha \mp \beta)}{2}$
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{i\theta} = A(\cos(\theta) + i\ \sin(\theta))$
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$

Reflection	Reflection phase shift
Off lower index	0
Off higher index	0.5 wavelength